COSH: Closed-Form Onset Search via Hyperbolic Response

Methodology for pre-flight critical-moment identification of a multirotor from ground-contact tip-over excitation

This document specifies the complete identification methodology, from raw signals to the take-off feed-forward moment. Every stage mirrors the implementation in `critical_value_getter_piecewise.py`; the companion analyses (model-fidelity bounds, excitation design, ground-effect bracket) are documented in `fidelity_section.tex` and are only referenced here.

1 Problem setting and notation

A multirotor rests on its landing gear. Ramping the commanded roll (pitch) moment tilts the vehicle until it begins to rotate about the landing-gear contact line — the *pivot*. The applied moment at that *rotation onset* is the critical moment M_{crit}; comparing the two tip directions yields the CoM offset and the feed-forward moment M_{ff} that compensates take-off. No rig, no motion capture and no stable in-flight data are used.

Symbol	Meaning	Value	Provenance
W, z_{CoM}	weight, CoM height	31.59 N, ∼0.3 m	measured / conservative
$l_{p,\phi}$, $l_{p,\theta}$	pivot offsets (roll, pitch)	0.140, 0.110 m	conservative geometry
J_P	inertia about the pivot line	—	never used numerically
C_T	thrust map, $T_i = C_T\Omega_i^2$	1.3175×10^{-7}	bench calibration
M_{thr}	window threshold	0.01 N m	noise floor
M_{cap}	allocator caps (x, y)	2.37, 2.74 N m	allocation feasibility
$\varepsilon_{\max}$	ramp-slope gate	3%	sensitivity chain (Sec. 4)
$N_{\min}$	window-sample gate	38	30 pre + 8 post
$e_{\text{lin,max}}$	linearity-RMSE gate	30 mN m	10× noise floor
N_{seg}	sweep margin (each end)	8	degenerate-candidate guard

Table 1: Design constants. (C_2, K) are *calibrated*, not set (Sec. 6).

2 Contact-phase dynamics and the closed form

About the active pivot line, with the small-angle attitude dependence retained ($\cos\varphi \approx 1$, $\sin\varphi \approx \varphi$), the roll/pitch dynamics read

$$J_P\,\dot\omega = M(t) + f\,l_p - W\,(l_p + \lambda_{\text{off}}) + W z_{\text{CoM}}\,\varphi, \tag{1}$$

where M is the commanded body moment, f the collective thrust and λ_{off} the CoM offset along the excited axis. Below the critical value the vehicle is static. At the onset the net moment vanishes, which supplies *two* anchoring conditions,

$$\omega(t_{\text{crit}}) = 0, \qquad \dot\omega(t_{\text{crit}}) = 0. \tag{2}$$

Past the onset gravity feeds back positively: the linearized dynamics have *real* eigenvalues $\pm C_2$ and, with the measured moment ramp $M(t) = M_{\text{crit}} + \dot M\,\tau$ in onset-relative time $\tau = t - t_{\text{crit}}$, the exact solution of (1)–(2) collapses to the hyperbolic closed form

$$\boxed{\omega(\tau) = C_1\big(\cosh(C_2\tau) - 1\big), \qquad C_1 = K\dot M, \quad K = \frac{1}{W z_{\text{CoM}}}, \quad C_2 = \sqrt{\frac{W z_{\text{CoM}}}{J_P}}} \tag{3}$$

for $\tau \geq 0$ (and $\omega = 0$ for $\tau < 0$). Because $J_P C_2^2 = W z_{\mathrm{CoM}}$, the amplitude C_1 is *fixed* by the measured ramp rate: with (C_2, K) pinned as rig constants and the baseline fixed by continuity, **no shape parameter is free per run**. The time-quadratic model of the original submission is the $C_2 \to 0$ limit of (3) — it discards the gravity term that creates the instability, with 14–75% truncation error over the windows actually used — and is retained only as a baseline.

3 Signals and excitation window

Per run, the moment is reconstructed from the measured rotor speeds, $M(t) = C_T \sum_i b_{a,i}\, \Omega_i^2(t)$, and $\omega(t)$ is taken from the odometry rate. The window end is $i_{\mathrm{end}} = \arg\max_t |M|$, *truncated at the first sample exceeding the allocator cap* M_{cap}: past it the thrust allocation saturates, the executed ramp is no longer linear, and the cosh family does not apply. (On the reference dataset the cap never binds; peak $|M| \leq 1.7$ N m.) The window start is the *last* sub-threshold sample before the peak, walking backwards — robust to spin-up blips that cross the threshold long before the actual ramp.

4 Onset-free run gates

Before any onset is estimated, each run must pass three gates evaluated on the measured moment trace alone (hence free of circularity):

1. **Ramp-slope error** $|\varepsilon| \leq 3\%$, with $\varepsilon = (|\dot{M}| - \dot{M}_{\mathrm{cmd}})/\dot{M}_{\mathrm{cmd}}$ and $\dot{M}$ the least-squares slope over the window. A 3% error in $\dot{M}$ perturbs $C_1 = K\dot{M}$ by 3% — about 7× below the ±20% level the sensitivity analysis shows harmless.

2. **Window samples** $N_{\mathrm{full}} \geq 38$: a necessary condition for any onset position. Because the window opens at $|M| = 10$ mN m while $M_{\mathrm{crit}} \gtrsim 0.4$ N m, the onset cannot occur earlier than $\hat{N}_{\mathrm{pre}} = M_{\mathrm{crit}}^{\mathrm{min}}/(\dot{M}\, T_s) \geq 30$ samples into the window even at the fastest ramp; adding the $N_{\mathrm{seg}} = 8$ post-onset fit minimum gives 38.

3. **Linearity RMSE** ≤ 30 mN m about the run's own best-fit line: a fault detector for aborted/stepped ramps, set at 10× the execution noise floor and above the healthy-run maximum.

The gates run *before* calibration so a poorly executed ramp cannot contaminate the dataset-coupled constraint $C_1 = K\dot{M}$.

5 Onset detection

With (C_2, K) pinned, detection reduces to an exhaustive sweep of the single remaining unknown — the onset index. Each candidate is a plain arithmetic evaluation: no optimizer, no initial guess, no seed model.

Algorithm 1 COSH onset detection (single run)

Require: samples $\{(t_i, \omega_i)\}$ on the excitation window $[i_{\mathrm{start}}, i_{\mathrm{end}}]$; moment $M(t)$; calibrated (C_2, K); measured ramp slope $\dot{M}$; margin $N_{\mathrm{seg}} = 8$
Ensure: onset index j^*; critical moment M_{crit}

$C_1 \leftarrow K\dot{M}$ ▷ onset conditions (2) fix the amplitude
for $j = i_{\mathrm{start}} + N_{\mathrm{seg}}, \ldots, i_{\mathrm{end}} - N_{\mathrm{seg}}$ **do** ▷ exhaustive sweep of the single unknown
  $C \leftarrow \mathrm{median}\{\omega_i : i_{\mathrm{start}} \leq i < j\}$ ▷ baseline by continuity
  $\mathrm{cost}(j) \leftarrow \sum_{i=i_{\mathrm{start}}}^{j-1} (\omega_i - C)^2 + \sum_{i=j}^{i_{\mathrm{end}}} \big(\omega_i - [\, C_1(\cosh C_2(t_i - t_j) - 1) + C\,]\big)^2$
end for
$j^* \leftarrow \arg\min_j \mathrm{cost}(j); \quad M_{\mathrm{crit}} \leftarrow M(t_{j^*})$

Two implementation details. *Baseline by median.* The pre-onset gyro trace is heavy-tailed (propeller vibration, spin-up spikes), and — more importantly — when a candidate j overshoots the true onset, the pre-segment tail contains the beginning of the hyperbolic rise. A mean baseline drifts with that leaked rise and partially cancels the residual it should expose, flattening the cost landscape; the median stays at the rest level until half the segment is contaminated, so misplaced candidates are penalized sharply and the minimum stays well defined. It is also a one-shot, seedless statistic, consistent with the no-optimizer design. An ablation rerunning the full pipeline — including the (C_2, K) calibration — with the mean instead (`analysis/baseline_stat_ablation.py`) leaves the run gating unchanged and 62/132 onsets identical, but biases the remainder toward *earlier* onsets: on the noisiest datasets the directional means shrink in magnitude by up to 30 mN m, precisely the leak mechanism above, while the ramp-invariance CV is essentially unchanged. The pivot-free offset M_{ff} moves by at most 4.2 mN m (0.13 mm), well below the run-to-run scatter — the median removes a systematic bias and tunes nothing. A second ablation pins $C = 0$, the ideal-sensor value implied by the onset conditions: the measured pre-onset baseline is in fact nonzero (gyro bias plus residual spin-up motion; $|C|$ median 3.1, p90 14.2, max 30.6 mrad/s), yet the estimate barely moves (71/132 onsets identical, M_{ff} shift ≤ 1.3 mN m). The estimated baseline is therefore not load-bearing on this dataset; it is kept because it makes the sweep invariant to the gyro bias — a rig- and temperature-dependent quantity that need not stay this small — at zero parameter cost. *Sweep margin.* The margin $N_{\mathrm{seg}} = 8$, applied at both window ends, guarantees every candidate split leaves at least N_{seg} samples in each segment; it excises only degenerate candidates: near $j = i_{\mathrm{start}}$ the baseline would be estimated from a handful of samples, and near $j = i_{\mathrm{end}}$ the post-onset arc is too short for the cosh branch to be falsifiable (at $j = i_{\mathrm{end}}$ its cost is nearly an empty sum). Since physically feasible onsets lie at least $\hat{N}_{\mathrm{pre}} \geq 30$ samples inside the window (Sec. 4) and the window closes only after the response is well developed, the sweep remains exhaustive over every feasible onset.

Remark 1 *Because the model curve is fully determined before the post-onset data are seen, goodness of fit is a* prediction test*, not flexibility: across all gate-passing runs the normalized post-onset residual is* $\approx 6\%$ *(gyro-noise dominated) and invariant over the twelvefold ramp-rate range — the empirical certificate that the response stays in the cosh family.*

6 Rig-constant calibration

Algorithm 1 presupposes (C_2, K). Although both have geometric expressions (3), they are *not* computed from geometry: z_{CoM} and especially J_P are not precisely known a priori, and — more importantly — the constants that make (3) reproduce the measured response are *effective* values that absorb the state-linear contributions of the unmodeled effects (the local slope of the gravity nonlinearity and the ground-effect spring). Both are calibrated once per dataset.

Joint residual minimization is not a usable criterion: in the cost of Algorithm 1, an earlier onset with a smaller amplitude trades off against a later onset with a larger one, so the summed residual is nearly degenerate along a ridge in the (C_2, K) plane and its minimizer is not physically meaningful. We calibrate instead by *physical self-consistency*: a static threshold must not depend on how fast it was approached. With runs at ramp rates $r \in \{0.10, \ldots, 1.20\}$ N m/s,

$$\mathrm{score}(C_2, K) \;=\; \sum_{\mathrm{dir} \in \{+,-\}} \mathrm{CV}\Big(\big\{M_{\mathrm{crit}}^{(r)}(C_2, K)\big\}_r\Big), \qquad \mathrm{CV} = \mathrm{std}/|\mathrm{mean}|, \tag{4}$$

where $M_{\mathrm{crit}}^{(r)}$ is the output of Algorithm 1 on the run with commanded rate r.

Search ranges. The admissible box is set so that it covers every physically plausible constant with margin. For $K = 1/(W z_{\mathrm{CoM}})$, the coarse range $K \in [0.10, 0.50]$ $(\mathrm{N\,m})^{-1}$ corresponds to $z_{\mathrm{CoM}} \in [0.06, 0.32]$ m at the vehicle weight — bracketing the plausible CoM heights — with the refinement allowed to move within the wider clip box $[0.05, 0.70]$ (z_{CoM} up to 0.63 m). For $C_2 \in [3, 8]$ rad/s, the implied pivot inertia $J_P = 1/(K C_2^2)$ spans roughly $[0.03, 1.1]$ kg m^2, bracketing the parallel-axis estimate $J + m z_{\mathrm{CoM}}^2 \approx 0.2$–$0.35$ kg m^2 by a factor of several on both sides.

Coarse-to-fine procedure. Algorithm 2 states the consistency criterion as a stand-alone routine — it wraps Algorithm 1 and returns the directional sum of coefficients of variation — and Algorithm 3 performs the search over it. The coarse scan evaluates (4) on an 11×11 grid ($\Delta C_2 = 0.5$ rad/s, $\Delta K = 0.04$ $(\mathrm{N\,m})^{-1}$; 121 evaluations). A single refinement pass then re-scans a 9×9 uniform grid spanning $\pm$ one coarse step about the coarse minimizer (± 0.5 rad/s, ± 0.04 $(\mathrm{N\,m})^{-1}$, clipped to the box), i.e. final resolutions of 0.125 rad/s and 0.01 $(\mathrm{N\,m})^{-1}$ — so the reported constants are not quantized to the coarse step. One pass suffices because the objective is flat along the (C_2, K) ridge; it is also *piecewise constant* (the onset is an integer index, so small parameter changes leave M_{crit} unchanged), which is why gradient-based or simplex optimizers stall on it, and a population-based global search (differential evolution) was found

Algorithm 2 Ramp-invariance consistency score

Require: gated runs at rates $r \in \{0.10, \ldots, 1.20\}$ N m/s, both tip directions; candidate pair (C_2, K)
Ensure: consistency score s ▷ lower = M_{crit} more rate-invariant
for each gated run r **do**
  $M_{\text{crit}}^{(r)} \leftarrow$ Algorithm 1 with (C_2, K) pinned ▷ stride-3 sweep
end for
for dir $\in \{+, -\}$ **do**
  $\mathrm{CV}_{\text{dir}} \leftarrow \mathrm{std}\big(\{M_{\text{crit}}^{(r)}\}_{r\in\text{dir}}\big) \,/\, \big|\mathrm{mean}\big(\{M_{\text{crit}}^{(r)}\}_{r\in\text{dir}}\big)\big|$
end for
return $s \leftarrow \mathrm{CV}_+ + \mathrm{CV}_-$ ▷ Eq. (4): a static threshold must not depend on the approach rate

Algorithm 3 Rig-constant calibration (coarse-to-fine)

Require: gated runs; box $C_2 \in [3, 8]$ rad/s, $K \in [0.05, 0.70]$ $(\mathrm{N\,m})^{-1}$
Ensure: calibrated pair (C_2^*, K^*)
coarse: evaluate Algorithm 2 on the 11×11 grid $C_2 = 3.0\!:\!0.5\!:\!8.0$, $K = 0.10\!:\!0.04\!:\!0.50$
$(C_2^b, K^b) \leftarrow \arg\min$ over the coarse grid
fine: evaluate Algorithm 2 on a 9×9 uniform grid over $[C_2^b \pm 0.5] \times [K^b \pm 0.04]$, clipped to the box ▷ resolution 0.125 rad/s, 0.01 $(\mathrm{N\,m})^{-1}$
$(C_2^*, K^*) \leftarrow \arg\min$ over the fine grid

to cost about 4× more for no measurable gain. During calibration the onset sweep of Algorithm 1 is subsampled by a stride of 3 samples for speed; the final identification runs at full resolution with the selected pair. Each evaluation is an arithmetic sweep, so no iterative optimizer is involved anywhere in the pipeline.

Remark 2 *The objective (4) retains a shallow valley, so the individual constants are only loosely determined: $Wz_{\text{CoM}} = 1/K$ and $J_P = 1/(KC_2^2)$ are order-of-magnitude sanity checks, not precision measurements. What the criterion pins down — by construction — is the identified onset: M_{crit} is robust along the valley, and a ±20% mis-specification of either constant moves the identified CoM by less than 0.9 mm.*

7 Aggregation and deliverables

Directional means over the gated runs give $\bar{M}_+$ and $\bar{M}_-$. The take-off feed-forward moment is the *pivot-free average*

$$M_{\text{ff}} = \tfrac{1}{2}\,(\bar{M}_+ + \bar{M}_-), \tag{5}$$

in which the pivot-arm terms $\pm(W - f)\,l_p$ — and any thrust-proportional contamination acting through the arm l_p, ground effect included — cancel by antisymmetry, while landing-gear asymmetry is deliberately *retained*: the same contact-phase moment acts at lift-off, where M_{ff} compensates it. The CoM offset follows from the moment-to-offset conversion of M_{ff} and is used for validation against load-cell ground truth only; M_{ff} itself is the quantity applied in flight.

8 Reproducibility

Every number in this document is reproducible from the repository: `critical_value_getter_piecewise.py` (pipeline, gates, caps, calibration, Algorithm 1); `analysis/ramp_quality.py` (gates); `analysis/sensitivity.py` (±20% robustness); `analysis/cosh_fidelity.py` (prediction-test residuals); `analysis/excitation_angle_design.py` (rate/cutoff design); `analysis/ge_linearity.py`, `ge_trajectory.py` (ground-effect bracket); `analysis/tiltcap_ablati` (window ablation).